# Prompt Library of local LLM test

***Domains: Lens Design | Photonics | Thermal Photonics | Quantum Photonics***

**This is a collection of test prompts for evaluating local LLMs for 4 specific domains:**

**(A) Lens Design – 5 tests**

**(B) Photonics – 4 tests**

**(C) Thermal Photonics – 3 tests**

**(D) Quantum Photonics – 4 tests**

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## HOW TO USE THIS DOCUMENT

1. Each test prompt is **self-contained** — copy and paste directly into LM Studio or AnythingLLM.
2. Every prompt includes a **preamble** that sets context, followed by the **actual task**.
3. **Reference answers** (where applicable) are provided after each prompt in a separate block — do NOT include these when testing the LLM.
4. **Scoring rubric** is provided at the end of each test.
5. Record: (a) model name/size, (b) platform, (c) quantization level, (d) full response, (e) time to respond.

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## SERIES A: TRADITIONAL LENS DESIGN

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### TEST A1: Paraxial Ray Tracing Through a Cemented Doublet

**Prompt**

| |
|---|
| You are an optical engineer. I will give you a cemented doublet lens prescription.<br><br>Your task is to: |

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(1) Trace a paraxial marginal ray from infinity (object at infinity, entrance pupil diameter = 20 mm) through the system.

(2) Trace a paraxial chief ray at a half-field angle of 10 degrees.

(3) Compute the following first-order properties:

- Effective Focal Length (EFL)
- Back Focal Length (BFL)
- Front Focal Length (FFL)
- Positions of the front and rear principal planes (H and H')
- f-number

Lens Prescription (all dimensions in mm):

Surface 1 (front of crown glass):

Radius of curvature: R1 = +61.47 mm

Surface 2 (cemented interface):

Radius of curvature: R2 = -43.32 mm

Thickness after Surface 1: t1 = 8.0 mm (BK7, nd = 1.5168, Vd = 64.17)

Surface 3 (rear of flint glass):

Radius of curvature: R3 = -124.60 mm

Thickness after Surface 2: t2 = 3.0 mm (SF2, nd = 1.6477, Vd = 33.85)

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After Surface 3: air (n = 1.0)

Please show ALL intermediate steps:

- Refraction and transfer equations used

- Ray heights and angles at each surface

- Sign convention you are using (state it explicitly before starting)

Present your results in a clearly organized table.

Do NOT use any external libraries or software. Work through the math step by step.

## Reference Answer Notes (DO NOT include in test)

- EFL should be approximately 99.7 mm
- Sign convention: Most common is the "real-is-positive" convention where distances measured left-to-right are positive, radii with centers to the right are positive
- Key failure modes: sign errors at cemented surface, forgetting to use n/n' correctly at refraction

### Scoring Rubric (10 points)

| Criterion | Points | What to Check |
|---|---|---|
| Correct sign convention stated | 1 | Must explicitly state before computing |
| Refraction equation correct | 2 | n*u* + *y*phi = n'*u' or equivalent |

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| Criterion | Points | What to Check |
|---|---|---|
| Transfer equation correct | 1 | y2 = y1 + t*u' (with correct n) |
| Marginal ray traced correctly | 2 | Heights and angles at each surface |
| Chief ray traced correctly | 1 | Correct field angle handling |
| EFL within 2% of reference | 2 | ~99.7 mm |
| BFL, FFL, principal planes computed | 1 | All four quantities present |

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## TEST A2: Seidel Aberration Coefficients for a Cooke Triplet

### Prompt

You are an expert in optical aberration theory. I will provide a Cooke triplet lens prescription. Your task is to compute the five Seidel (third-order) aberration coefficients for this system.

Cooke Triplet Prescription (all dimensions in mm, wavelength = 587.56 nm):

| Surf | Radius(mm) | Thickness(mm) | Glass |
|---|---|---|---|
| 1 | +26.30 | 3.80 | SK16 (nd=1.6204, Vd=60.30) |
| 2 | +141.80 | 6.44 | Air |
| 3 | -48.90 | 1.00 | F2 (nd=1.6200, Vd=36.37) |
| 4 | +24.91 | 5.28 | Air |
| 5 | +103.70 | 3.50 | SK16 (nd=1.6204, Vd=60.30) |
| 6 | -30.80 | --- | Air |

System specs:

- f-number: f/3.5

- EFL: approximately 50 mm

- Object at infinity

- Entrance pupil at Surface 1

Tasks:

(1) First, trace the paraxial marginal ray and chief ray (half-field = 15 degrees).

(2) At each surface, compute the Seidel aberration contributions:

- S_I (Spherical Aberration)

- S_II (Coma)

- S_III (Astigmatism)

- S_IV (Petzval Curvature)

- S_V (Distortion)

(3) Sum the contributions to get total system Seidel sums.

(4) Present a surface-by-surface breakdown table.

(5) Discuss which surfaces are the dominant contributors to each aberration,

and explain physically why.

Use the standard Seidel formulation:

S_I = -A^2 * y * delta(u/n)

(or equivalent — state your formulation explicitly)

where A = n*i = n'*i' is the refraction invariant.

**Scoring Rubric (10 points)**

| Criterion | Points | What to Check |
|---|---|---|
| Correct Seidel formulation stated | 1 | Must show which convention/textbook |
| Marginal ray correctly traced | 2 | Required for Seidel computation |
| Chief ray correctly traced | 1 | Required for off-axis aberrations |
| Surface-by-surface table present | 2 | All 6 surfaces x 5 aberrations |
| Seidel sums physically reasonable | 2 | Signs and magnitudes make sense |
| Physical discussion of dominant surfaces | 2 | Why certain surfaces dominate |

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## TEST A3: Merit Function Construction for an Imaging Lens

**Prompt**

You are a senior lens designer working in Code V. A client has the following requirements for a machine vision imaging lens:

Specifications:

- Focal length: 25 mm +/- 0.5 mm

- f-number: f/2.0

- Full field of view: 60 degrees (half-field = 30 degrees)

- Spectral range: 450 nm to 650 nm (visible)

- Image format: 1/1.8" sensor (7.2 mm diagonal)

- Performance target: MTF > 40% at 100 lp/mm across the full field

- Distortion: < 1% at maximum field

- Relative illumination: > 70% at maximum field

- Total track length: < 35 mm

- Working distance (BFL): > 10 mm

- Operating temperature: -10 C to +60 C

Tasks:

(1) Design a merit function for optimizing this lens. Specify:

- Which field points to sample (and why)

- Which wavelengths to use and their weights

- What aberration operands or ray targets to include

- How to weight center field vs edge field

- Boundary constraints (thickness, edge thickness, clearances)

- How to handle the distortion and relative illumination constraints

(2) Explain your weighting strategy. Why did you choose these weights?

What trade-offs are you making?

(3) If you were using Code V, write the corresponding merit function in

Code V macro language (or pseudo-code that closely resembles it).

(4) Describe your optimization strategy:

- What variables would you release first?
- Would you use DLS, global optimization, or a staged approach?
- How would you handle the thermal requirement?

(5) What starting configuration would you recommend?

(e.g., double Gauss, Tessar, retrofocus — justify your choice)

**Scoring Rubric (10 points)**

| Criterion | Points | What to Check |
|---|---|---|
| Appropriate field sampling (3+ points) | 1 | On-axis, 0.7 field, full field minimum |
| Wavelength selection with weights | 1 | At least 3 wavelengths, photopic weighting |
| Operand selection makes physical sense | 2 | Ray aberrations or wavefront, not random |
| Weighting strategy explained | 2 | Center vs edge, sagittal vs tangential |
| Boundary constraints included | 1 | Thickness, clearance, total track |
| Optimization strategy is practical | 1 | Staged approach preferred |
| Starting configuration justified | 1 | Double Gauss is most appropriate here |

| Criterion | Points | What to Check |
|---|---|---|
| Code V syntax approximately correct | 1 | Basic structure recognizable |

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## TEST A4: Glass Selection for an Apochromatic Objective

**Prompt**

You are a lens designer tasked with selecting glasses for an apochromatic (APO) objective lens. The requirements are:

- EFL: 100 mm
- f/4
- Spectral range: 435.8 nm (g-line) to 656.3 nm (C-line)
- Must achieve apochromatic correction (three wavelengths brought to common focus)
- Prefer commercially available glasses (Schott or Ohara catalogs)
- Budget is moderate — avoid exotic crystals like CaF2 if possible

Tasks:

(1) Explain the difference between achromatic and apochromatic correction in terms of the secondary spectrum.

(2) Explain why a standard crown-flint achromat cannot achieve apochromatic correction. Use the concept of partial dispersion (P) and the P-V diagram.

(3) Identify at least TWO candidate glass combinations (triplet) that can achieve apochromatic correction. For each combination:

- List the glass names (e.g., S-FPL51, S-BSM81)
- Provide nd, Vd, and Pg,F (partial dispersion) for each glass
- Explain why this combination works on the P-V diagram
- Discuss pros and cons (cost, availability, chemical durability, thermal sensitivity)

(4) For your preferred glass combination, sketch the lens configuration (how many elements, which glass goes where, cemented or air-spaced) and explain the reasoning.

(5) Discuss at least TWO practical considerations:

- Thermal sensitivity (dn/dT, CTE)
- Manufacturability (stain resistance, softness)
- Cost and lead time
- Environmental compliance (RoHS — lead-free, arsenic-free)

**Scoring Rubric (10 points)**

| Criterion | Points | What to Check |
|---|---|---|
| Achromat vs. apochromat explained correctly | 1 | Secondary spectrum concept |

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| Criterion | Points | What to Check |
|---|---|---|
| P-V diagram concept correct | 2 | Normal line, anomalous dispersion |
| At least 2 valid glass combinations | 2 | Must use real glass names with correct properties |
| Glass properties approximately correct | 2 | nd, Vd, Pg,F within reasonable range |
| Lens configuration makes sense | 1 | Triplet with correct power distribution |
| Practical considerations discussed | 2 | At least 2 of thermal, manufacturing, cost, RoHS |

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## TEST A5: Python Code Generation — Complete Ray Tracing Engine

**Prompt**

```
Write a complete, standalone Python script that implements a sequential

ray tracing engine for rotationally symmetric optical systems.


Requirements:

(1) The script must be COMPLETE and RUNNABLE — no placeholders, no "..."

(2) Use only standard libraries plus numpy and matplotlib

(3) All characters must be ASCII-compatible (no Greek letters, no Unicode symbols)

(4) Include proper file encoding: use encoding='utf-8' for all file I/O


The ray tracer must support:

- Spherical surfaces (defined by radius of curvature)
```

- Flat surfaces (radius = infinity)

- Refraction using Snell's law (not paraxial approximation)

- Sequential tracing through user-defined surface list

- Both on-axis and off-axis rays (finite field angles)

The script should:

(a) Define a lens system as a list of surfaces, each with:

- Radius of curvature (mm)

- Thickness to next surface (mm)

- Refractive index after the surface

(b) Implement real (non-paraxial) ray tracing:

- Input: ray height y and angle u at first surface

- Trace through all surfaces using exact Snell's law

- Handle total internal reflection (detect and report)

(c) As a built-in test case, define the following cemented doublet:

Surface 1: R = +61.47 mm, t = 8.0 mm, n = 1.5168 (BK7)

Surface 2: R = -43.32 mm, t = 3.0 mm, n = 1.6477 (SF2)

Surface 3: R = -124.60 mm, t = (to image)

Entrance pupil diameter: 20 mm

(d) Trace a fan of rays (at least 11 rays across the pupil) for:

- On-axis (0 degrees)
- 5 degrees off-axis
- 10 degrees off-axis

(e) Generate and save the following plots:

- Ray fan diagram (transverse ray aberration vs pupil coordinate)
- Spot diagram at the image plane for each field angle
- 2D layout drawing showing the lens cross-section and ray paths

(f) Save all plots as PNG files with descriptive filenames

(g) Print a summary table of:

- Paraxial EFL, BFL
- RMS spot radius at each field angle
- Maximum ray aberration at each field angle

Structure the code with clear functions:

- trace_ray(surfaces, y, u) -> list of (y, u) at each surface
- spot_diagram(surfaces, field_angle, num_rays) -> (x_coords, y_coords)
- ray_fan(surfaces, field_angle, num_rays) -> (pupil_coords, aberrations)
- draw_layout(surfaces, rays) -> matplotlib figure

Include docstrings for each function.

Add a if __name__ == "__main__": block that runs the full analysis.

**Scoring Rubric (10 points)**

| Criterion | Points | What to Check |
|---|---|---|
| Code is complete and runnable | 2 | No missing functions, no placeholders |
| Snell's law correctly implemented | 2 | Exact, not paraxial |
| Test case lens correctly defined | 1 | Matches prescription given |
| Ray fan diagram generated | 1 | Correct axes, meaningful curves |
| Spot diagram generated | 1 | Points at image plane |
| Layout drawing generated | 1 | Lens cross-section visible |
| ASCII-only, no encoding issues | 1 | No Unicode in strings or comments |
| Code is modular and reusable | 1 | Functions with docstrings, clean structure |

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## SERIES B: PHOTONICS

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### TEST B1: Waveguide Mode Solving — Silicon Photonics

**Prompt**

You are a photonics engineer working on silicon photonics at 1550 nm wavelength.

Consider a silicon ridge waveguide with the following parameters:

- Core material: Silicon (n_Si = 3.476 at 1550 nm)

- Cladding material: SiO2 (n_SiO2 = 1.444 at 1550 nm)

- Waveguide width: w = 500 nm

- Waveguide height (total silicon thickness): h = 220 nm

- Slab height (partially etched): h_slab = 0 nm (i.e., fully etched —

  this is actually a channel/strip waveguide)

- Top cladding: Air (n = 1.0) [asymmetric cladding]

Tasks:

(1) Determine whether this waveguide supports single-mode or multi-mode

operation at 1550 nm. Explain your reasoning using the V-number concept

adapted for rectangular waveguides, or effective index method.

(2) Using the effective index method (EIM):

- First solve the 1D slab waveguide in the vertical (y) direction

  to find the effective index of the fundamental slab mode

- Then solve the 1D slab waveguide in the horizontal (x) direction

  using the effective index from step 1

- Report the effective index (n_eff) for the fundamental TE mode (TE00)

- Report the effective index for the fundamental TM mode (TM00)

(3) Show all equations used:

- Transcendental equation for the symmetric/asymmetric slab waveguide

- How you solve it (graphical method, Newton-Raphson, etc.)

(4) Discuss the accuracy limitations of the effective index method compared to a full vectorial mode solver (e.g., finite difference eigenmode, or FEM).

(5) Calculate the following derived quantities:

- Group index (n_g) at 1550 nm (you may assume dn_Si/d_lambda = -1.0e-4 nm^-1 near 1550 nm)
- Confinement factor (approximate, using the overlap integral concept)
- Effective mode area (A_eff, approximate)

(6) If I increase the waveguide width to 600 nm, would a second TE mode appear? Estimate the cutoff width for the TE01 mode.

**Scoring Rubric (10 points)**

| Criterion | Points | What to Check |
|---|---|---|
| Single-mode determination correct | 1 | 500x220 nm Si is single-mode at 1550 nm |
| EIM vertical solve correct | 2 | Asymmetric slab, correct equation |
| EIM horizontal solve correct | 1 | Using n_eff from vertical as core index |
| n_eff for TE00 reasonable | 2 | Should be roughly 2.3-2.5 |
| TE vs TM difference | 1 | TM has lower n_eff due to field orientation |

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| Criterion | Points | What to Check |
|---|---|---|
| explained | | |
| EIM limitations discussed | 1 | Corner effects, polarization coupling |
| Cutoff width estimate reasonable | 1 | ~700-800 nm for TE01 |
| Group index calculation | 1 | Correct use of n_g = n_eff - lambda * dn_eff/dlambda |

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## TEST B2: Coupled Mode Theory — Directional Coupler

### Prompt

You are designing a directional coupler for a silicon photonics chip operating

at 1550 nm.

Coupler parameters:

- Two identical silicon strip waveguides (500 nm wide, 220 nm tall)

- SiO2 lower cladding, air upper cladding

- Gap between waveguides: g = 200 nm

- Effective index of isolated waveguide mode: n_eff = 2.44 (TE polarization)

- Operating wavelength: lambda = 1550 nm

Tasks:

(1) Using coupled mode theory (CMT), derive the expressions for:

- The even (symmetric) and odd (anti-symmetric) supermodes

- The coupling coefficient kappa in terms of the supermode indices

- The coupling length L_c = pi / (2 * kappa)

- The power transfer ratio as a function of coupler length L:

  P_cross / P_input = sin^2(kappa * L)

(2) Estimate the coupling coefficient kappa for the given geometry.

You may use the approximation that kappa decays exponentially with gap:

  kappa ~ kappa_0 * exp(-gamma * g)

where gamma is the evanescent decay constant in the gap region.

Calculate gamma from the waveguide parameters.

(3) Compute the coupling length L_c for a 3-dB coupler (50:50 split).

(4) Discuss the wavelength dependence of the coupling ratio. If the coupler

is designed for 50:50 at 1550 nm, what is the approximate splitting ratio

at 1530 nm and 1570 nm? (Assume kappa changes linearly with wavelength.)

(5) Write the 2x2 transfer matrix for this directional coupler:

```
[E_through]     [cos(kappa*L)          -j*sin(kappa*L)] [E_in1]
[E_cross    ] = [-j*sin(kappa*L)     cos(kappa*L)     ] [E_in2]
```

Verify that this matrix is unitary.

(6) If I want to build a wavelength-insensitive coupler, what design modifications would you suggest? Discuss at least two approaches.

**Scoring Rubric (10 points)**

| Criterion | Points | What to Check |
|---|---|---|
| CMT derivation correct | 2 | Supermode concept, coupling coefficient |
| Coupling coefficient estimation reasonable | 2 | Exponential decay, gamma calculation |
| L_c for 3dB coupler correct | 1 | L_c = pi/(2*kappa), reasonable value |
| Wavelength dependence discussed | 2 | Qualitative trend correct |
| Transfer matrix correct and unitary | 1 | Phase factors correct |
| Wavelength-insensitive approaches | 2 | MZI coupler, adiabatic coupler, or broadband MMI |

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## TEST B3: FDTD Simulation Setup — Photonic Crystal

**Prompt**

Write a complete, standalone Python script that simulates electromagnetic wave propagation through a 2D photonic crystal using the Finite-Difference Time-Domain (FDTD) method.

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Requirements:

- Use ONLY numpy and matplotlib (no MEEP, no external EM libraries)

- All characters must be ASCII-compatible (no Greek letters, no Unicode)

- Include encoding='utf-8' for all file I/O

- The script must be COMPLETE and RUNNABLE

Photonic Crystal Parameters:

- 2D square lattice of air holes in dielectric (like a silicon slab)

- Lattice constant: a = 400 nm

- Hole radius: r = 0.3 * a = 120 nm

- Background dielectric constant: epsilon_r = 12.0 (silicon)

- Hole dielectric constant: epsilon_r = 1.0 (air)

- Simulation domain: at least 10a x 10a

- TM polarization (Ez, Hx, Hy)

FDTD Implementation:

(a) Implement the Yee grid for 2D TM mode:

    - Update equations for Ez, Hx, Hy

    - Courant stability condition: dt <= dx / (c * sqrt(2))

    - Grid resolution: at least 20 points per lattice constant

(b) Source: Gaussian pulse point source at the center-left of the domain

    - Center frequency corresponding to a/lambda = 0.3

- Bandwidth wide enough to cover a/lambda = 0.2 to 0.5

(c) Boundary conditions:

- Implement Mur's first-order absorbing boundary conditions

  (or simple PML if you prefer, but Mur's ABC is acceptable)

(d) Create the photonic crystal structure:

- Function to generate the dielectric map (epsilon on the grid)
- Air holes arranged in square lattice

(e) Run the simulation for enough time steps to see the pulse

propagate through the crystal (at least 1000 steps)

(f) Output:

- Plot 1: Dielectric constant map (showing the photonic crystal)
- Plot 2: Ez field snapshot at several time steps (e.g., t = 200, 500, 1000)
- Plot 3: Time signal recorded at a monitor point on the far side
- Save all plots as PNG files

(g) Print simulation parameters: grid size, time step, total time,

number of grid points, memory usage estimate

Structure with clear functions:

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```
- create_photonic_crystal(Nx, Ny, a, r, eps_bg, eps_hole, dx)

- update_fields(Ez, Hx, Hy, epsilon, dx, dt)

- apply_boundary(Ez, Hx, Hy, ...)

- gaussian_source(t, t0, sigma, freq)
```

### Scoring Rubric (10 points)

| Criterion | Points | What to Check |
|---|---|---|
| Yee grid update equations correct | 3 | Curl equations discretized properly |
| Courant condition enforced | 1 | dt calculated correctly |
| Photonic crystal geometry correct | 1 | Square lattice of holes |
| Source implementation correct | 1 | Gaussian pulse, correct frequency |
| Boundary conditions present | 1 | Mur ABC or PML, not just hard walls |
| Code runs without errors | 1 | Complete, no missing functions |
| Output plots generated | 1 | At least dielectric map + field snapshot |
| ASCII-only encoding | 1 | No Unicode characters |

---

## TEST B4: Photonic Device Design — Mach-Zehnder Modulator

### Prompt

```
You are a photonic device engineer designing a Mach-Zehnder modulator (MZM)
```

on a thin-film lithium niobate (TFLN) platform.

Platform specifications:

- Substrate: Lithium niobate thin film (X-cut, 600 nm thick LN on SiO2)

- Waveguide type: Rib waveguide, etched 300 nm, width 1.0 um

- Electrode material: Gold

- Operating wavelength: 1550 nm

- Target modulation bandwidth: > 40 GHz

- Electro-optic coefficient: r33 = 31 pm/V (for LN)

- Extraordinary index: n_e = 2.138 at 1550 nm

Tasks:

(1) Calculate V_pi * L (the voltage-length product) for this modulator.

Show the derivation starting from the electro-optic effect:

- Phase shift: delta_phi = (pi / lambda) * n_e^3 * r33 * Gamma * (V/d) * L

- where Gamma is the electro-optic overlap integral, d is the electrode gap

- V_pi * L = lambda * d / (n_e^3 * r33 * Gamma)

Assume:

- Electrode gap: d = 5 um

- Electro-optic overlap integral: Gamma = 0.5

(2) For a target V_pi = 3.5 V, calculate the required modulator length L.

(3) Design the electrode structure for 40 GHz bandwidth:

- What electrode geometry (CPW — coplanar waveguide) would you use?
- Discuss velocity matching between the RF wave and optical wave
- What is the required RF effective index (n_RF)?
- How does the electrode gap affect both V_pi and bandwidth?
- Discuss impedance matching to 50 ohms

(4) Calculate the 3-dB electro-optic bandwidth considering:

- Velocity mismatch: f_3dB_vel = c / (pi * L * |n_RF - n_g_opt|)
- RC bandwidth limitation
- Which is the dominant bandwidth limitation for your design?

(5) Discuss the trade-offs:

- V_pi vs. bandwidth (shorter L -> higher V_pi but wider bandwidth)
- Electrode gap vs. V_pi vs. impedance
- Insertion loss budget (propagation loss, coupling loss, splitting loss)

(6) Compare TFLN MZM performance against:

- Silicon MZM (carrier depletion, plasma dispersion)
- InP MZM (quantum-confined Stark effect)
- Provide a comparison table with V_pi*L, bandwidth, loss, and footprint

(7) If the client asks for a modulator with V_pi < 2V and bandwidth > 60 GHz,

what design modifications would you propose? Is this feasible on TFLN?

**Scoring Rubric (10 points)**

| Criterion | Points | What to Check |
|---|---|---|
| V_pi*L derivation correct | 2 | Starting from electro-optic effect |
| V_pi*L numerical value reasonable | 1 | Should be ~2-3 V*cm for TFLN |
| Modulator length for V_pi=3.5V correct | 1 | Consistent with V_pi*L |
| Velocity matching concept correct | 2 | n_RF must match n_g_opt |
| Bandwidth calculation reasonable | 1 | >40 GHz achievable with short L |
| Trade-off discussion substantive | 1 | Not just listing — actual analysis |
| Platform comparison table provided | 1 | At least Si, InP, TFLN |
| Advanced design modifications discussed | 1 | Segmented electrodes, capacitively loaded, etc. |

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**SERIES C: THERMAL PHOTONICS**

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## TEST C1: Planck Radiation and Selective Thermal Emission

### Prompt

You are a thermal radiation engineer working on thermophotovoltaic (TPV)

energy conversion.

Tasks:

(1) Write the complete Planck spectral radiance formula:

B(lambda, T) = (2*h*c^2 / lambda^5) * 1 / (exp(h*c/(lambda*k_B*T)) - 1)

Define every variable and its SI units.

(2) For a blackbody at T = 1500 K:

- Calculate the peak emission wavelength using Wien's displacement law
- Calculate the total emissive power using the Stefan-Boltzmann law
- Calculate the spectral radiance at lambda = 2.0 um (in W/m^2/sr/um)

(3) Now consider a SELECTIVE EMITTER designed for a GaSb TPV cell

(bandgap E_g = 0.73 eV, corresponding to lambda_g = 1.70 um):

Ideal selective emitter properties:

- Emissivity epsilon = 0.95 for lambda < 1.70 um (above bandgap)
- Emissivity epsilon = 0.05 for lambda > 1.70 um (below bandgap)

Calculate:

- The useful radiated power (lambda < 1.70 um) per unit area

- The wasted radiated power (lambda > 1.70 um) per unit area

- The spectral efficiency: eta = P_useful / P_total

- Compare with the spectral efficiency of a blackbody

(4) Explain Kirchhoff's law of thermal radiation:

- State the law precisely

- Under what conditions does it apply?

- What does it imply for the design of selective emitters?

- How does it connect absorptivity and emissivity?

(5) Discuss at least THREE physical mechanisms for engineering

selective thermal emission:

- Photonic crystals

- Metamaterials / metasurfaces

- Rare-earth oxides (e.g., erbium oxide)

- Multilayer thin-film stacks

For each, explain the physical principle and give one example

from the literature.

(6) Write a Python script that:

- Plots Planck spectral radiance at 1500 K for lambda = 0.5 to 10 um

- Overlays the selective emitter spectrum (epsilon * B(lambda, T))
- Shades the "useful" and "wasted" spectral regions
- Computes and prints the spectral efficiency
- Saves the plot as PNG
- Uses only numpy and matplotlib
- ASCII-only characters, encoding='utf-8' for file I/O

**Scoring Rubric (10 points)**

| Criterion | Points | What to Check |
|---|---|---|
| Planck formula correct with units | 1 | All variables defined |
| Wien peak wavelength correct (~1.93 um) | 1 | Wien constant = 2898 um*K |
| Stefan-Boltzmann total power correct | 1 | sigma*T^4 |
| Selective emitter efficiency calculated | 2 | Integration approach correct |
| Kirchhoff's law stated correctly | 1 | Spectral, directional form |
| Three emission engineering mechanisms | 2 | Physical principles explained |
| Python code runs and produces correct plot | 2 | Planck curve shape, shading, efficiency |

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## TEST C2: Transfer Matrix Method for Multilayer Thermal Emitter

**Prompt**

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***Domains: Lens Design | Photonics | Thermal Photonics | Quantum Photonics***

Write a complete, standalone Python script that designs a narrowband

thermal emitter using the Transfer Matrix Method (TMM).

The emitter is a 1D multilayer stack on a metallic substrate.

Design target:

- Narrowband emission peak centered at lambda = 4.0 um

  (mid-infrared, for CO2 gas sensing application)

- Peak emissivity > 0.9

- FWHM bandwidth < 0.5 um

- Low emissivity (< 0.1) outside the emission band (2-8 um range)

Stack structure (initial design — your code should also allow modification):

- Substrate: Tungsten (use Drude model or tabulated optical constants)

- Layer 1: SiO2 (n ~ 1.40 in mid-IR, but include dispersion if possible)

- Layer 2: Si (n ~ 3.42 in mid-IR)

- Layer 3: SiO2

- (repeat as needed for Fabry-Perot cavity design)

Requirements:

(1) Implement the full Transfer Matrix Method:

  - 2x2 transfer matrix for each layer

  - Interface matrices (Fresnel coefficients)

- Propagation matrices

- Handle both TE and TM polarizations

- Compute reflectance R and transmittance T

- Compute absorptance/emittance: A = 1 - R - T

  (for opaque substrate, T = 0, so emittance = 1 - R)

(2) Use the Drude model for tungsten:

- epsilon(omega) = 1 - omega_p^2 / (omega^2 + j*omega*gamma)

- omega_p = 1.22e16 rad/s (plasma frequency)

- gamma = 6.71e13 rad/s (damping rate)

- Compute n and k from epsilon

(3) Optimize layer thicknesses to achieve the target emission spectrum:

- Start with quarter-wave thicknesses at 4.0 um

- Implement a simple optimization (scipy.optimize or manual sweep)

- Target: maximize emissivity at 4.0 um while minimizing out-of-band

(4) Generate plots:

- Emissivity vs wavelength (2-8 um) for the optimized design

- Emissivity vs wavelength for TE and TM at normal incidence

- Emissivity vs angle (0-80 degrees) at lambda = 4.0 um

- Save all plots as PNG

(5) Print:

- Optimized layer thicknesses
- Peak emissivity and peak wavelength
- FWHM of emission peak
- Average out-of-band emissivity

Code requirements:

- Complete and standalone
- numpy, matplotlib, scipy only
- ASCII-only characters
- encoding='utf-8' for all file I/O
- Modular functions with docstrings
- if __name__ == "__main__": block

**Scoring Rubric (10 points)**

| Criterion | Points | What to Check |
|---|---|---|
| TMM implementation correct | 3 | Interface + propagation matrices, both polarizations |
| Drude model for tungsten correct | 1 | Complex epsilon to n,k conversion |
| Emissivity = 1 - R (opaque substrate) | 1 | Correctly handled |
| Optimization implemented | 2 | Some method to adjust thicknesses |

| Criterion | Points | What to Check |
|---|---|---|
| Angular dependence computed | 1 | TE and TM at multiple angles |
| Code runs and produces meaningful results | 1 | Peak near 4 um, narrow band |
| ASCII-only, proper encoding | 1 | No Unicode |

---

## TEST C3: Daytime Radiative Cooling Design

### Prompt

```
You are a researcher designing a passive daytime radiative cooling surface

that can achieve sub-ambient cooling without any energy input.


Background:

- The atmosphere has a transparency window between approximately 8-13 um

- A surface can cool by radiating thermal energy through this window into

    cold outer space (~3 K)

- During daytime, the surface must also reflect >95% of solar radiation

    (0.3-2.5 um) to avoid solar heating


Tasks:

(1) Energy balance analysis:

        Write the complete energy balance equation for a radiative cooler:


        P_cool = P_rad(T) - P_atm(T_amb) - P_solar - P_conduction - P_convection
```

where:

- P_rad = integral of epsilon(lambda,theta) * B(lambda,T) over all wavelengths, angles, and hemisphere
- P_atm = integral of epsilon(lambda,theta) * epsilon_atm(lambda,theta) * B(lambda,T_amb) over all wavelengths and angles
- P_solar = integral of (1-rho(lambda)) * I_solar(lambda) d_lambda

Estimate each term numerically for:

- Surface temperature T = 300 K (ambient)
- Ambient temperature T_amb = 300 K
- Solar irradiance: 1000 W/m^2 (AM1.5)

What is the maximum achievable cooling power at ambient temperature?

What is the theoretical minimum temperature below ambient?

(2) Material and structure design:

Propose a practical multilayer structure for daytime radiative cooling.

Specify:

- Layer materials and thicknesses
- Why each layer is chosen (solar reflector, IR emitter, substrate)
- Expected spectral properties

Consider at least TWO approaches:

(a) Polymer-based design (e.g., PDMS, PMMA, PE films)

(b) Multilayer photonic design (e.g., HfO2/SiO2 stacks)

(3) Atmospheric transmission model:

- Explain which molecules cause absorption in the 8-13 um window (O3, H2O, CO2)
- How does humidity affect cooling performance?
- Provide approximate atmospheric transmittance values in the 8-13 um window for dry vs humid conditions

(4) Write a Python script that:

- Models the atmospheric transmittance in the 8-13 um window (use a simplified model with Gaussian absorption bands for O3 at 9.6 um and CO2 at 15 um)
- Computes the net cooling power as a function of surface temperature (from 250 K to 310 K) for an ideal selective emitter
- Plots: cooling power vs temperature, showing equilibrium point
- Plots: spectral comparison of surface emission vs atmospheric window
- Saves all plots as PNG
- Uses only numpy and matplotlib
- ASCII-only, encoding='utf-8'

(5) Discuss practical challenges:

- Dust and dirt accumulation
- Wind and convective heat gain
- Scalability and cost
- How to integrate with building HVAC systems

(6) What is the current state-of-the-art in sub-ambient cooling?

What temperature drops below ambient have been demonstrated?

What are the main research frontiers?

**Scoring Rubric (10 points)**

| Criterion | Points | What to Check |
|---|---|---|
| Energy balance equation complete | 2 | All terms present and explained |
| Numerical estimates reasonable | 1 | ~100 W/m^2 max cooling at ambient |
| Two material approaches proposed | 2 | Polymer-based AND photonic design |
| Atmospheric window explained | 1 | 8-13 um, O3/H2O/CO2 effects |
| Python code runs correctly | 2 | Cooling power curve, spectral plot |
| Practical challenges discussed | 1 | At least 3 real-world issues |
| State-of-the-art knowledge | 1 | ~5-10 C below ambient demonstrated |

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## SERIES D: QUANTUM PHOTONICS

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### TEST D1: Single-Photon Source — g(2) Measurement

**Prompt**

You are a quantum optics experimentalist characterizing a single-photon source.

Background:

A quantum dot (QD) single-photon source emitting at 920 nm is being

characterized using a Hanbury Brown-Twiss (HBT) interferometer setup:

- 50:50 beamsplitter

- Two single-photon avalanche detectors (SPADs)

- Time-correlated single-photon counting (TCSPC) electronics

- Time resolution (detector jitter): 350 ps

- Dark count rate per detector: 100 counts/s

- Repetition rate of pulsed excitation: 80 MHz

- Average detected count rate: 50,000 counts/s per detector

Tasks:

(1) Explain the second-order correlation function g^(2)(tau):

- Write the mathematical definition

- What does g^(2)(0) = 0 mean physically?

- What values indicate: single-photon source, coherent source,

thermal/bunched source?

- Why is g^(2)(0) < 0.5 the commonly accepted threshold for single-photon emission?

(2) Describe the HBT experiment:

- Draw (describe in text) the experimental setup
- How is the coincidence histogram built from detector clicks?
- What is the role of the time delay tau?
- How do you normalize the histogram to get g^(2)(tau)?

(3) Data analysis:

Given the following coincidence histogram data (counts in each time bin around tau = 0, bin width = 1 ns, symmetric):

tau (ns): -10, -9, -8, ..., -2, -1, 0, 1, 2, ..., 8, 9, 10

Counts at tau = 0 peak: 12

Counts at tau = +/- 12.5 ns: 245 (first side peak, = 1/rep_rate)

Counts at tau = +/- 25 ns: 250 (second side peak)

Counts at tau = +/- 37.5 ns: 248 (third side peak)

Background (between peaks): 5 counts per bin

Calculate:

- Raw g^(2)(0) = (area of zero-delay peak) / (average area of side peaks)
- Background-corrected g^(2)(0) using:

  g^(2)_corrected = (g^(2)_raw - (1-rho^2)) / rho^2

  where rho = S / (S + B), S = signal rate, B = background rate
- Estimate the multi-photon probability

(4) Write a Python script that:

- Simulates photon arrival times from a pulsed single-photon source

  with g^(2)(0) = 0.05 (small multi-photon probability)
- Simulates the HBT experiment: beamsplitter, two detectors,

  coincidence counting
- Includes detector dark counts and timing jitter (Gaussian, 350 ps)
- Builds and plots the coincidence histogram
- Computes g^(2)(0) from the histogram
- Saves the plot as PNG
- Uses only numpy and matplotlib
- ASCII-only, encoding='utf-8'

(5) Discuss practical limitations:

- How does detector jitter affect the measured g^(2)(0)?
- How do dark counts affect it?
- What about detector dead time and afterpulsing?
- How would you improve the measurement?

### Scoring Rubric (10 points)

| Criterion | Points | What to Check |
|---|---|---|
| g^(2)(tau) definition correct | 2 | Intensity correlation, normalization |
| Physical meaning of g^(2)(0) values correct | 1 | 0=single, 1=coherent, >1=bunched |
| HBT setup described correctly | 1 | Beamsplitter + 2 detectors + coincidence |
| g^(2)(0) calculated from given data | 2 | Raw and background-corrected |
| Python simulation runs correctly | 2 | Generates histogram with dip at tau=0 |
| Practical limitations discussed | 2 | Jitter, dark counts, dead time |

---

## TEST D2: Entangled Photon Pairs — SPDC Phase Matching

### Prompt

You are a quantum optics researcher working with spontaneous parametric

down-conversion (SPDC) for generating entangled photon pairs.

Setup:

- Nonlinear crystal: Beta-Barium Borate (BBO)

- Pump wavelength: lambda_p = 405 nm (blue diode laser)

- Type-II phase matching (signal and idler have orthogonal polarizations)

- Crystal cut angle to be determined

- Goal: Generate degenerate photon pairs at 810 nm

BBO Properties:

- Negative uniaxial crystal (n_e < n_o)

- Sellmeier equations for BBO:

n_o^2 = 2.7359 + 0.01878 / (lambda^2 - 0.01822) - 0.01354 * lambda^2

n_e^2 = 2.3753 + 0.01224 / (lambda^2 - 0.01667) - 0.01516 * lambda^2

where lambda is in micrometers.

Tasks:

(1) Derive the phase-matching conditions for Type-II SPDC:

- Energy conservation: omega_p = omega_s + omega_i

- Momentum conservation (phase matching): k_p = k_s + k_i

- For collinear Type-II: n_p^e(theta) * / lambda_p =

n_s^o / lambda_s + n_i^e(theta) / lambda_i

Write the expression for the extraordinary index as a function

of angle theta:

1/n_e^2(theta) = cos^2(theta)/n_o^2 + sin^2(theta)/n_e^2

(2) For degenerate SPDC (lambda_s = lambda_i = 810 nm):

- Calculate n_o and n_e for BBO at 405 nm and 810 nm using the Sellmeier equations above
- Determine the phase-matching angle theta_pm
- Show your calculation step by step

(3) Discuss non-collinear phase matching:

- What happens when you allow the signal and idler to propagate at angles to the pump?
- Draw (describe in text) the emission cone geometry
- At what cone half-angle do the signal and idler cones intersect?
- Why are these intersection points useful for polarization entanglement?

(4) Calculate the SPDC bandwidth:

- The spectral bandwidth is determined by the phase-matching bandwidth
- For a crystal length L = 2 mm, estimate the FWHM bandwidth of the down-converted photons
- Use: delta_lambda ~ 0.44 * lambda_s^2 / (L * |dn_s/dlambda - dn_i/dlambda|)

(5) Write a Python script that:

- Implements the BBO Sellmeier equations
- Computes n_o and n_e vs wavelength (300-1000 nm)
- Finds the Type-II phase-matching angle for degenerate 810 nm SPDC
- Plots the phase-matching curve: signal wavelength vs crystal angle

- Plots the SPDC emission spectrum (sinc^2 function) for L = 2 mm
- Saves all plots as PNG
- Uses only numpy and matplotlib
- ASCII-only, encoding='utf-8'

(6) Discuss practical considerations:

- Spatial and spectral filtering of SPDC photons
- Walk-off compensation (how and why?)
- Pair generation rate and brightness
- Crystal length trade-offs (brightness vs bandwidth vs walk-off)

**Scoring Rubric (10 points)**

| Criterion | Points | What to Check |
|---|---|---|
| Phase-matching conditions correct | 2 | Energy + momentum conservation |
| Sellmeier equations applied correctly | 1 | Correct n_o, n_e at 405 and 810 nm |
| Phase-matching angle reasonable | 2 | ~42-43 degrees for degenerate Type-II BBO |
| Non-collinear geometry described | 1 | Cone picture, intersection points |
| SPDC bandwidth estimated | 1 | Order of magnitude correct (~5-10 nm) |

| Criterion | Points | What to Check |
|---|---|---|
| Python code runs correctly | 2 | Phase-matching curve, spectrum |
| Walk-off and practical issues | 1 | Compensation crystal, filtering |

---

## TEST D3: Quantum Key Distribution — BB84 Protocol Simulation

### Prompt

You are a quantum information scientist implementing and analyzing the

BB84 quantum key distribution protocol.

Tasks:

(1) Explain the BB84 protocol step by step:

- Alice's random bit generation and basis selection
- Encoding in quantum states (rectilinear and diagonal bases)
- Bob's random basis selection and measurement
- Public basis comparison (sifting)
- Error estimation (QBER)
- Privacy amplification (conceptual level)

(2) Security analysis:

- Explain why an eavesdropper (Eve) necessarily introduces errors
- If Eve uses the intercept-resend attack, what is the expected QBER?
- What is the maximum tolerable QBER for secure key generation?
- Briefly discuss the difference between individual attacks,

collective attacks, and coherent attacks

(3) Realistic channel model:

Consider a fiber-based QKD system with:

- Fiber loss: 0.2 dB/km (standard SMF-28 at 1550 nm)

- Fiber length: L = 50 km

- Single-photon detector efficiency: eta_det = 15% (InGaAs SPAD)

- Dark count rate: R_dark = 1e-6 per gate (per detection window)

- Gate width: 1 ns

- Source repetition rate: 1 GHz

- Optical misalignment error: e_opt = 1%

Calculate:

- Channel transmittance: eta_ch = 10^(-alpha*L/10)

- Total detection probability per pulse: eta_total = eta_ch * eta_det

- Signal count rate

- Dark count rate contribution to QBER

- Total QBER

- Secure key rate using the GLLP formula (simplified):

R_key = q * Q_mu * [1 - H(e_mu) - f*H(e_mu)]

where q=1/2 (for BB84), Q_mu is the gain, e_mu is the QBER,

H is binary entropy, f~1.16 is error correction efficiency

(4) Decoy state protocol:

- Explain why weak coherent pulses (WCP) are vulnerable to photon number splitting (PNS) attacks
- How does the decoy state method solve this?
- If the mean photon number is mu = 0.5, what fraction of pulses contain 0, 1, 2, 3+ photons? (Poisson statistics)

(5) Write a complete Python script that simulates the BB84 protocol:

The simulation should:

(a) Generate N = 100,000 random bits for Alice

(b) Alice randomly selects rectilinear (+) or diagonal (x) basis for each bit

(c) Simulate Bob's random basis selection

(d) For matching bases, Bob's measurement outcome equals Alice's bit (in the absence of noise)

(e) Add realistic noise:

- Channel loss (each photon has probability eta_ch * eta_det of arriving)
- Dark counts (probability p_dark of a click even without a photon)
- Optical error (probability e_opt of a bit flip)

(f) Simulate an optional eavesdropper (Eve) with intercept-resend attack:

- Eve randomly chooses a basis, measures, resends
- This introduces ~25% QBER on intercepted bits
- Eve intercepts a fraction f_eve of the pulses

(g) Perform basis sifting

(h) Estimate QBER from a random sample (10% of sifted key)

(i) Compute the secure key rate

(j) Print a summary: raw key length, sifted key length, QBER,

estimated secure key rate, whether key generation is secure

(k) Generate plots:

- QBER vs eavesdropping fraction (f_eve from 0 to 1)
- Secure key rate vs fiber length (0 to 200 km)
- Save as PNG

Requirements:

- Complete, standalone, runnable
- numpy and matplotlib only
- ASCII-only, encoding='utf-8'

(6) If the QBER exceeds 11%, what should Alice and Bob conclude?

What are their options?

**Scoring Rubric (10 points)**

| Criterion | Points | What to Check |
|---|---|---|
| BB84 steps explained correctly | 1 | All steps from encoding to sifting |

**Prompt Library of local LLM test**

***Domains: Lens Design | Photonics | Thermal Photonics | Quantum Photonics***

| Criterion | Points | What to Check |
|---|---|---|
| Intercept-resend QBER = 25% | 1 | Correct derivation |
| Channel loss calculation correct | 1 | 10 dB for 50 km, eta_ch correct |
| QBER calculation complete | 2 | Signal + dark count contributions |
| Decoy state concept correct | 1 | PNS attack and how decoy solves it |
| Python simulation runs | 2 | BB84 protocol simulated end-to-end |
| Key rate vs distance plot correct | 1 | Rate drops, reaches zero ~100-150 km |
| QBER > 11% interpretation correct | 1 | Abort, Eve detected |

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**TEST D4: Quantum Sensing with Squeezed Light**

**Prompt**

You are a quantum sensing researcher working on gravitational wave

detection using squeezed light injection.

Context:

LIGO-type Michelson interferometer with Fabry-Perot arm cavities.

The goal is to improve phase sensitivity beyond the standard quantum

limit (SQL) using squeezed vacuum states.

Tasks:

(1) Standard Quantum Limit (SQL):

- Derive the phase sensitivity of a standard Michelson interferometer

  operating at the dark port with coherent light input:

  delta_phi_SQL = 1 / sqrt(N)

where N is the average photon number.

- Explain the two noise sources that contribute to the SQL:

  (a) Shot noise (photon counting noise)

  (b) Radiation pressure noise

- At what power level do these two contributions balance?

- Why can't you simply increase laser power to improve sensitivity

  indefinitely?

(2) Squeezed states of light:

- Define a squeezed vacuum state |0, r> where r is the squeeze parameter

- Write the quadrature variances:

  delta X1 = (1/2) * exp(-r)  (squeezed quadrature)

  delta X2 = (1/2) * exp(+r)  (anti-squeezed quadrature)

- Verify that the uncertainty product satisfies:

  delta_X1 * delta_X2 = 1/4 (minimum uncertainty)

- What is the squeeze factor in dB? Express as: S_dB = -10*log10(exp(-2r))

- For r = 1.15 (corresponding to 10 dB squeezing), what are the

  quadrature variances?

(3) Squeezed light injection into the interferometer:

- Explain how squeezed vacuum is injected at the dark port
- Which quadrature should be squeezed to reduce shot noise?
- Derive the improved phase sensitivity:

  delta_phi = exp(-r) / sqrt(N)

- For 10 dB of squeezing, what is the improvement factor?

(4) Frequency-dependent squeezing:

- Explain why simple squeezing only helps at high frequencies

  (where shot noise dominates)

- At low frequencies, radiation pressure noise dominates and

  the anti-squeezed quadrature makes things WORSE

- How does frequency-dependent squeezing (using a filter cavity)

  solve this problem?

- What is the required filter cavity length and linewidth for

  LIGO-like parameters?

(5) Numerical calculation:

Consider a simplified interferometer:

- Arm length: L = 4 km
- Laser wavelength: lambda = 1064 nm
- Laser power at beamsplitter: P = 750 kW (after power recycling)
- Mirror mass: M = 40 kg

- Squeeze parameter: r = 1.15 (10 dB)

Calculate:

- Average photon number per measurement interval (tau = 1 s)
- SQL phase sensitivity: delta_phi_SQL
- Improved phase sensitivity with squeezing: delta_phi_squeezed
- Corresponding strain sensitivity: h = delta_phi * lambda / (4*pi*L)
- At what frequency does radiation pressure noise equal shot noise?

(6) Write a Python script that:

- Computes the noise spectral density of an interferometer with

  and without squeezed light injection
- Plots: strain sensitivity vs frequency (10 Hz to 10 kHz)

  showing four curves:

  (a) Shot noise limit (no squeezing)

  (b) Radiation pressure noise limit

  (c) Total noise (SQL)

  (d) Total noise with 10 dB squeezing (frequency-independent)

  (e) Total noise with 10 dB frequency-dependent squeezing
- Use log-log axes
- Save the plot as PNG
- numpy and matplotlib only
- ASCII-only, encoding='utf-8'

(7) Advanced discussion:

- What is the Heisenberg limit and how does it differ from the SQL?
- Can squeezing alone reach the Heisenberg limit? Why or why not?
- What quantum states would be needed to reach the Heisenberg limit?
- Discuss the current state of squeezed light in LIGO/Virgo/KAGRA.

**Scoring Rubric (10 points)**

| Criterion | Points | What to Check |
|---|---|---|
| SQL derivation correct | 2 | Shot noise + radiation pressure |
| Squeezed state variances correct | 1 | exp(-r), exp(+r), uncertainty product |
| Squeeze injection concept correct | 1 | Dark port, correct quadrature |
| Phase sensitivity improvement correct | 1 | Factor of exp(-r) |
| Frequency-dependent squeezing explained | 2 | Filter cavity concept |
| Numerical calculations reasonable | 1 | Order-of-magnitude strain sensitivity |
| Python code produces correct curves | 1 | SQL, shot, RP, squeezed curves |
| Heisenberg limit discussion | 1 | 1/N scaling, NOON states or similar |

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**Qualitative Assessment Checklist**

## Prompt Library of local LLM test

***Domains: Lens Design | Photonics | Thermal Photonics | Quantum Photonics***

For each test, also record (Yes/No/Partial):

| **Quality Metric** | **Description** |
|---|---|
| Correct physics? | Are the physical principles and equations correct? |
| Correct numerics? | Are numerical answers within 5% of reference? |
| Code runs? | Does generated Python code execute without errors? |
| Code gives correct output? | Does the code produce physically correct results? |
| Explains reasoning? | Does the model show its work and explain why? |
| Identifies limitations? | Does it mention what it cannot do or where it is uncertain? |
| Design insight? | Does it show engineering judgment beyond textbook formulas? |
| Appropriate depth? | Is the response at the right technical level? |

## Platform Comparison Notes

| **Aspect** | **LM Studio** | **AnythingLLM** |
|---|---|---|
| Model loaded | | |
| Quantization | | |
| Context length used | | |
| Response time (s) | | |
| RAG documents loaded | N/A | (list docs) |
| RAG improvement observed? | N/A | Yes/No/Marginal |

**Prompt Library of local LLM test**

***Domains: Lens Design | Photonics | Thermal Photonics | Quantum Photonics***

| Aspect | LM Studio | AnythingLLM |
|---|---|---|
| Interface notes | | |